

CES and Isoelastic Forms

Equivalence, Conversion, and Partial Adjustment

Technical Note

1. Two ways of writing the same behaviour

There are two natural ways to model how fuel consumption responds to prices. They look different but, for small price changes, they describe the same thing.

The CES form

The CES (constant elasticity of substitution) is a *structural* model. It starts from a cost-minimisation problem: a sector chooses a bundle of fuels to deliver energy services at minimum cost, subject to a CES aggregator. The solution gives fuel quantities as a function of all prices simultaneously:

$$q_f = q_f^0 \left(\frac{p_f}{p_f^0} \right)^{-\sigma} \left(\frac{C}{C^0} \right)^{\sigma+\eta}$$

where C is the CES cost index (a function of *all* fuel prices), σ is the substitution elasticity, and η is the demand elasticity. The key feature is that C couples all fuels together: changing the price of one fuel affects the quantities of every other fuel through the cost index.

The isoelastic form

The isoelastic (constant elasticity) model is a *reduced-form* description. It writes the response of each fuel directly in terms of own-price and cross-price elasticities:

$$\frac{\Delta q_f}{q_f} \approx \varepsilon_{ff} \frac{\Delta p_f}{p_f} + \sum_{g \neq f} \varepsilon_{fg} \frac{\Delta p_g}{p_g}$$

where ε_{ff} is the own-price elasticity and ε_{fg} are cross-price elasticities. This is a first-order log-linear approximation—it says each fuel responds to every price with a constant elasticity, and the total response is additive.

Why they are equivalent (locally)

The CES *generates* a specific pattern of isoelastic elasticities. Taking the log-differential of the CES quantity equation and using the properties of the CES cost index, one obtains:

$$\varepsilon_{ff} = -\sigma(1 - s_f) + (\sigma + \eta)s_f = -\sigma + (\sigma + \eta)s_f \quad (1)$$

$$\varepsilon_{fg} = (\sigma + \eta)s_g \quad \text{for } g \neq f \quad (2)$$

where s_f is the *cost share* of fuel f (not the energy share). Equation (1) decomposes the own-price elasticity into a substitution effect ($-\sigma$, always negative) and a cost-index effect $((\sigma + \eta)s_f$, positive if $\eta > -\sigma$). Equation (2) says all cross-price elasticities are proportional to the other fuel's cost share—this is the CES's characteristic restriction.

So the CES is a *disciplined* isoelastic model: it generates an entire $n \times n$ elasticity matrix from just two parameters (σ and η) plus the observed cost shares. An unconstrained isoelastic model would require n^2 parameters. The CES achieves parsimony by imposing:

- Symmetry: the cross-elasticity of f with respect to g is related to the cross-elasticity of g with respect to f via cost shares.

- Proportionality: all cross-elasticities from a given fuel g are proportional to s_g .
- Adding up: the elasticities satisfy Engel aggregation and Cournot aggregation automatically.

For small price changes (say, under 20% of the base price), the isoelastic approximation to the CES is very accurate. For large price changes, the two forms diverge because the CES properly tracks the nonlinear reallocation of shares, while the isoelastic form doesn't update shares as prices change.

When to use which

	CES	Isoelastic
Spreadsheet implementation	Needs cost index C	Direct per-fuel formulas
Large price changes	Accurate	Approximation degrades
Consistency of cross-elasticities	Guaranteed	Must be imposed manually
Number of parameters	2+ shares	n^2 (or 2+ shares if from CES)
Analytical tractability	Cost function available	Simpler algebra
Partial adjustment	Natural	Natural

For the effective carbon price formula, we use the isoelastic (semi-elasticity) form because we are linearising around the baseline. This is exact for small policy perturbations and avoids the need to compute the cost index C inside the formula itself. But the elasticities we plug in come from the CES, so the two forms are working in concert: the CES provides the discipline, the isoelastic form provides the simplicity.

2. Partial adjustment

Both forms describe *long-run equilibrium* responses—the quantities you would observe if the economy had fully adjusted to the new prices. In practice, adjustment takes time: capital stocks must turn over, infrastructure must be built, habits must change. A partial adjustment model captures this.

The idea

In any year t , actual quantities don't jump to the equilibrium. Instead, they move a fraction λ of the way from last year's actual quantity toward this year's equilibrium target:

$$q_{f,t} = q_{f,t-1} + \lambda (q_{f,t}^* - q_{f,t-1})$$

where $q_{f,t}^*$ is the equilibrium quantity implied by year- t prices and $\lambda \in (0, 1]$ is the adjustment speed. This is equivalent to an exponential smoothing filter on the equilibrium path, with half-life $\approx \ln 2 / \lambda$.

The parameter λ creates a distinction between short-run and long-run elasticities. If the long-run substitution elasticity is σ , then the effective elasticity in any single year is approximately $\lambda\sigma$ for the first year after a price change, rising toward σ over time as the adjustment accumulates.

Partial adjustment works identically in both forms

The partial adjustment equation operates on *quantities*, not on the price-response model that generates the equilibrium target. So it sits as a layer on top of whichever form you use:

1. Compute the equilibrium target $q_{f,t}^*$ from year- t prices using *either* the CES form or the isoelastic form.
2. Apply the partial adjustment filter to get actual $q_{f,t}$.

The only difference is how step 1 is computed:

In the CES form, the equilibrium target is computed relative to the previous year’s *actual* quantities and prices (not the original baseline), using the full CES equations. This means the cost index C must be recomputed each year from actual prices, and the quantity response reflects the current fuel mix, not the original one. The CES automatically handles the fact that as shares evolve, the substitution possibilities change.

In the isoelastic form, the equilibrium target is also computed relative to the previous year’s actuals, using the elasticity matrix. But since the elasticities are constant (they don’t update with shares), the model doesn’t capture the changing substitution landscape as the fuel mix evolves. For year-to-year steps this error is small, but it accumulates over long horizons.

This is the one substantive difference partial adjustment introduces between the two forms: the CES *re-linearises* at each year’s actual position, while the isoelastic form uses a fixed linearisation. For a 10-year projection with moderate price changes, the difference is typically small. For transformational scenarios (e.g., near-complete electrification), the CES is substantially more reliable.

Adjustment speed

The parameter λ can vary by:

- **Sector:** power generation adjusts faster than buildings (shorter capital cycles).
- **Fuel pair:** switching between gasoline and diesel is fast (λ near 1); switching from gas to electricity involves new infrastructure (λ perhaps 0.1).
- **Margin:** demand reduction (extensive) may adjust faster than fuel switching (intensive) in some sectors, or vice versa, justifying separate λ^{int} and λ^{ext} .

If different λ values are used for the intensive and extensive margins, the effective carbon price formula still applies in any given year—one simply uses the partially-adjusted quantities (which reflect the different speeds) rather than the equilibrium quantities when computing emissions reductions.

Appendix: Formula Reference

Throughout: fuel f in sector s , year t . Baseline (year 0) quantities q_{fs}^0 , prices p_{fs}^0 , emission factors e_f . CES substitution elasticity σ_s , demand elasticity η_s , adjustment speed λ_s (or λ_s^{int} , λ_s^{ext} if split by margin).

A.1 CES form — equilibrium quantities

CES cost index at time t :

$$C_{s,t} = \left(\sum_f \alpha_{fs} p_{fs,t}^{1-\sigma_s} \right)^{1/(1-\sigma_s)} \quad (3)$$

where α_{fs} are calibrated share parameters from the base year.

Equilibrium quantity of fuel f at time t , relative to previous year's actuals:

$$q_{fs,t}^{\text{CES}} = q_{fs,t-1} \left(\frac{p_{fs,t}}{p_{fs,t-1}} \right)^{-\sigma_s} \left(\frac{C_{s,t}}{C_{s,t-1}} \right)^{\sigma_s + \eta_s} \quad (4)$$

A.2 Isoelastic form — equilibrium quantities

Equilibrium quantity of fuel f at time t , relative to previous year's actuals:

$$q_{fs,t}^{\text{iso}} = q_{fs,t-1} \prod_g \left(\frac{p_{gs,t}}{p_{gs,t-1}} \right)^{\varepsilon_{fg,s}} \quad (5)$$

where the elasticities are derived from the CES:

$$\varepsilon_{ff,s} = -\sigma_s + (\sigma_s + \eta_s) s_{fs} \quad (6)$$

$$\varepsilon_{fg,s} = (\sigma_s + \eta_s) s_{gs} \quad (g \neq f) \quad (7)$$

and $s_{fs} = p_{fs} q_{fs} / \sum_g p_{gs} q_{gs}$ is the cost share.

Note: In the CES form, the cost shares s_{fs} evolve endogenously through C . In the isoelastic form, they are held fixed at the linearisation point. This is the source of divergence over long horizons.

A.3 Partial adjustment — applied to either form

Single adjustment speed:

$$q_{fs,t} = q_{fs,t-1} + \lambda_s (q_{fs,t}^* - q_{fs,t-1}) \quad (8)$$

where $q_{fs,t}^*$ is from either A.1 (CES) or A.2 (isoelastic).

Effective short-run elasticity in year 1 after a price shock: $\approx \lambda_s \sigma_s$ (intensive) and $\approx \lambda_s |\eta_s|$ (extensive).

Half-life of adjustment: $t_{1/2} = \ln 2 / \ln(1/(1 - \lambda_s)) \approx \ln 2 / \lambda_s$ for small λ_s .

Split adjustment speeds (intensive and extensive margins adjust at different rates):

Decompose the equilibrium into a fuel-mix component and a demand component. Define:

$$Q_{s,t}^* = \sum_f q_{fs,t}^* \quad (\text{equilibrium total energy in sector}) \quad (9)$$

$$w_{fs,t}^* = q_{fs,t}^* / Q_{s,t}^* \quad (\text{equilibrium energy shares}) \quad (10)$$

Then partially adjust each margin separately:

$$Q_{s,t} = Q_{s,t-1} + \lambda_s^{\text{ext}} (Q_{s,t}^* - Q_{s,t-1}) \quad (11)$$

$$w_{fs,t} = w_{fs,t-1} + \lambda_s^{\text{int}} (w_{fs,t}^* - w_{fs,t-1}) \quad (12)$$

Actual quantities:

$$\boxed{q_{fs,t} = w_{fs,t} \cdot Q_{s,t}} \quad (13)$$

This ensures shares sum to one at every time step (since equilibrium shares sum to one and partial adjustment of shares preserves the sum), while allowing fuel switching and demand reduction to proceed at different speeds.

A.4 Relationship between adjustment speed and elasticity horizon

After a permanent price change at $t = 0$, the cumulative adjustment by year T is:

$$\frac{q_{f,T} - q_{f,0}}{q_f^* - q_{f,0}} = 1 - (1 - \lambda)^T$$

So the *effective elasticity at horizon T* is:

$$\sigma_T^{\text{eff}} = \sigma [1 - (1 - \lambda)^T] \quad (14)$$

This provides a direct mapping between the partial adjustment model and the distinction between “short-run” and “long-run” elasticities reported in the empirical literature. If the literature reports σ^{SR} at a 1-year horizon and σ^{LR} at a long horizon, then:

$$\lambda \approx \frac{\sigma^{\text{SR}}}{\sigma^{\text{LR}}} \quad \text{and} \quad \sigma = \sigma^{\text{LR}} \quad (15)$$

The same relationship holds for η and λ^{ext} .